

Supplementary Data

lifelines-hc: a Python package for Higher Criticism testing in two-sample survival analysis under non-proportional hazards

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This document accompanies the Application Note *lifelines-hc: a Python package for Higher Criticism testing in two-sample survival analysis under non-proportional hazards*. It provides (i) the full analysis workflow and scripts used to reproduce the case studies, and (ii) complete per-method test statistics and p -values for all four datasets analysed in the main text.

1 Analysis scripts and data

All scripts live in the `AppNote/` directory of the `lifelines-hc` repository (<https://github.com/alonkipnis/lifelines-hc/tree/main/AppNote>). They require Python ≥ 3.8 , `lifelines-hc`, and `lifelines` ≥ 0.29 .

Workflow

1. **Download trial IPD.** `01_get_kmdata.py` exports reconstructed individual patient data (IPD) from the `kmdata` R package [Guyot et al., 2012] for CheckMate 057 [Borghaei et al., 2015], COMET-1 [Smith et al., 2016], and AZURE [Coleman et al., 2011, 2014]. CSL liver-cirrhosis IPD are downloaded directly from the CRAN `timereg` package and the `SurvSet` repository [Martinussen and Scheike, 2006, Drysdale, 2022] (see `run_csl.py`).
2. **Run per-dataset analyses.** Each `run_*.py` script loads one dataset, calls `run_all_tests()` in `utils.py`, writes a results CSV to `results/`, and saves diagnostic figures to `figs/`.
3. **Build the composite figure.** `make_figure.py` assembles Figure 1 of the main text from the four domain scripts.

Script index

Reproduction commands

```
cd AppNote
python 01_get_kmdata.py --trials Checkmate057_1C COMET1_2A AZURE_2A
python run_immuno_oncology.py
python run_comet.py
python run_azure.py
python run_csl.py           # requires SurvSet or AppNote/data/CSL.csv
python make_figure.py
```

Each `run_*.py` script writes `results/<domain>_test_results.csv` containing one row per competing method (statistic and permutation- or asymptotic-based p -value).

Table 1: Analysis scripts for the four case studies.

Script	Dataset	Role
<code>01_get_kmdata.py</code>	CheckMate 057, COMET-1, AZURE	Download IPD via <code>kmdata</code> (Guyot algorithm).
<code>run_immuno_oncology.py</code>	CheckMate 057 PFS ($n = 582$)	Nivolumab vs. docetaxel; crossing curves.
<code>run_comet.py</code>	COMET-1 OS ($n = 1028$)	Cabozantinib vs. prednisone; transient early benefit.
<code>run_azure.py</code>	AZURE DFS ($n = 3359$)	Zoledronic acid vs. control; menopause-dependent effect.
<code>run_csl.py</code>	CSL OS ($n = 446$)	Prednisone vs. placebo; public raw IPD, multi-window effect.
<code>utils.py</code>	—	Shared <code>run_all_tests()</code> , plotting, and table utilities.
<code>make_figure.py</code>	All four	Composite Figure 1 panels (A–H).

2 Data sources

Table 2 lists the original clinical studies, endpoints, and how each dataset was obtained. For CheckMate 057, COMET-1, and AZURE, only published Kaplan–Meier curves were available; IPD were reverse-engineered with the algorithm of Guyot et al. [2012] and distributed in the `kmdata` package (<https://github.com/raredd/kmdata>). The CSL trial is the only case study with genuinely public raw IPD.

Table 2: Original studies and data acquisition for the four case studies.

Trial	Endpoint	n	Source and acquisition
CheckMate 057	PFS	582	Borghaei et al. [2015]; IPD reconstructed from published Kaplan–Meier curves via Guyot et al. [2012] (<code>kmdata</code>).
COMET-1	OS	1028	Smith et al. [2016]; IPD reconstructed from published Kaplan–Meier curves via Guyot et al. [2012] (<code>kmdata</code>).
AZURE	DFS	3359	Coleman et al. [2011, 2014]; IPD reconstructed from published Kaplan–Meier curves via Guyot et al. [2012] (<code>kmdata</code>).
CSL	OS	446	Copenhagen liver-cirrhosis trial (1962–1969); raw IPD from the CRAN <code>timereg</code> package [Martinussen and Scheike, 2006] and <code>SurvSet</code> [Drysdale, 2022].

Reconstructed IPD files are written to `AppNote/data/` as `Checkmate057_1C.csv`, `COMET1_2A.csv`, and `AZURE_2A.csv` by `01_get_kmdata.py`. CSL data are written to `AppNote/data/CSL.csv` by `run_csl.py`.

3 Shared analysis settings

Unless noted otherwise, HCHG uses:

- equal-width time pooling with `n_intervals_to_pool` as listed per dataset below;
- two-sided interval hypergeometric p -values (`alternative='both'`);
- HC fraction $\gamma = 0.2$;
- label-permutation p -values with the random seed fixed at 42.

The log-rank test and four weighted log-rank variants (Gehan-Wilcoxon, Tarone-Ware, Peto-Prentice, Fleming-Harrington with $p = q = 1$) use the standard `lifelines.statistics.logrank_test` implementation with asymptotic p -values.

4 Per-method results tables

Tables 3–6 report the full comparison for each dataset. Asterisks mark $p < 0.05$. Permutation-based p -values for HCHG are discrete (multiples of $1/(n_{\text{perms}} + 1)$ with $n_{\text{perms}} = 2000$) and are rounded to three decimals; “ < 0.001 ” denotes the smallest attainable value, $1/2001$.

CheckMate 057 — progression-free survival ($n = 582$)

Original trial: [Borghaei et al. \[2015\]](#). IPD reconstructed from published Kaplan–Meier curves [[Guyot et al., 2012](#)]. Control: docetaxel ($n = 290$); treatment: nivolumab ($n = 292$). HC settings: $K = 60$ intervals, $n_{\text{perms}} = 2000$.

Table 3: CheckMate 057 PFS: all competing methods.

Method	Statistic	p -value
Log-rank	0.868	0.351
Gehan-Wilcoxon	4.169	0.041*
Tarone-Ware	0.844	0.358
Peto-Prentice	3.869	0.049*
Fleming-Harrington (1,1)	1.292	0.256
HCHG	1.695	0.001*

COMET-1 — overall survival ($n = 1028$)

Original trial: [Smith et al. \[2016\]](#). IPD reconstructed from published Kaplan–Meier curves [[Guyot et al., 2012](#)]. Control: prednisone ($n = 346$); treatment: cabozantinib ($n = 682$). HC settings: $K = 80$ intervals, $n_{\text{perms}} = 2000$.

AZURE — disease-free survival ($n = 3359$)

Original trial: [Coleman et al. \[2011, 2014\]](#). IPD reconstructed from published Kaplan–Meier curves [[Guyot et al., 2012](#)]. Control: standard adjuvant therapy ($n = 1678$); treatment: zoledronic acid ($n = 1681$). HC settings: $K = 80$ intervals, $n_{\text{perms}} = 2000$.

Table 4: COMET-1 OS: all competing methods.

Method	Statistic	p -value
Log-rank	1.259	0.262
Gehan-Wilcoxon	1.693	0.193
Tarone-Ware	1.570	0.210
Peto-Prentice	1.602	0.206
Fleming-Harrington (1,1)	0.648	0.421
HCHG	1.179	0.014*

Table 5: AZURE DFS: all competing methods.

Method	Statistic	p -value
Log-rank	1.054	0.305
Gehan-Wilcoxon	0.652	0.419
Tarone-Ware	0.691	0.406
Peto-Prentice	1.145	0.284
Fleming-Harrington (1,1)	0.277	0.598
HCHG	1.513	0.005*

CSL — overall survival ($n = 446$)

Original trial: Copenhagen Study Group for Liver Diseases (1962–1969); public IPD distributed via [Martinussen and Scheike \[2006\]](#) and [Drysdale \[2022\]](#). Control: placebo ($n = 220$); treatment: prednisone ($n = 226$). Follow-up times are in years. HC settings: $K = 50$ intervals, $n_{\text{perms}} = 2000$.

Table 6: CSL liver-cirrhosis OS: all competing methods.

Method	Statistic	p -value
Log-rank	0.757	0.384
Gehan-Wilcoxon	0.439	0.507
Tarone-Ware	0.774	0.379
Peto-Prentice	0.574	0.449
Fleming-Harrington (1,1)	2.234	0.135
HCHG	1.257	0.002*

5 Cross-dataset summary

Table 7 condenses the log-rank and HCHG p -values reported in the main text. Full statistics for all six methods appear in Section 4.

Across all four datasets, HCHG achieves $p \leq 0.014$ while the log-rank test is non-significant throughout ($p \geq 0.26$). Among the weighted log-rank alternatives, only CheckMate 057 Gehan-Wilcoxon and Peto-Prentice reach $p < 0.05$ (at 0.041 and 0.049 respectively); every other weighted test returns $p \geq 0.13$.

Table 7: Summary of log-rank and HCHG p -values across the four case studies (full tables in Section 4).

Dataset	Endpoint	n	Log-rank p	HCHG p
CheckMate 057	PFS	582	0.351	0.001
COMET-1	OS	1028	0.262	0.014
AZURE	DFS	3359	0.305	0.005
CSL	OS	446	0.384	0.002

6 Machine-readable results

The CSV files below contain the same values as Tables 3–6 (columns `method`, `statistic`, `p_value`):

- `AppNote/results/immuno_test_results.csv`
- `AppNote/results/comet_test_results.csv`
- `AppNote/results/azure_test_results.csv`
- `AppNote/results/csl_test_results.csv`

References

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- Torben Martinussen and Thomas H. Scheike. *Dynamic Regression Models for Survival Data*. Springer, New York, 2006. doi: 10.1007/0-387-33960-4. CSL liver cirrhosis trial (prednisone vs placebo, Copenhagen Study group for Liver diseases, 1962–1969); dataset `csl` in the CRAN `timereg` package.

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